

Teen Mental Health — Data Analysis

Exploratory analysis of "Cleaned_Teen_Mental_Health.csv" — methods: Python (Pandas, Matplotlib, Plotly). Presenter: [Your Name]





Why this matters

Adolescence is a critical period for mental-health development. Early signals in population data guide prevention, resource allocation and targeted interventions across schools and clinics.

Objectives

Describe population patterns, identify risk factors, surface actionable correlations, and produce evidence-based recommendations.

Approach

Clean, summarise and visualise features; compute correlations and produce interactive charts for exploration.

Dataset overview

Source file: Cleaned_Teen_Mental_Health.csv

Shape & features

Rows: — (use actual row count) ·

Columns: — (use actual column count)

- Age
- Gender
- Stress Level
- Sleep Hours
- Screen Time
- Physical Activity
- Anxiety
- Depression
- Academic Performance

Data types & notes

Mixed numeric and categorical types. Key cleaning points: consistent gender labels, numeric scaling for survey scores, datetime parsing where present.

	Date	Region	Product	Sales Rep	Units Sold	Unit Price	Total Sales	Discount %	Final Sales
1	1/2/2024	North	Laptop	Alice Johnson	12	\$1,200.00	\$14,400.00	5%	\$13,680.00
2	1/3/2024	South	Monitor	Bob Smith	24	\$300.00	\$7,200.00	10%	\$6,480.00
3	1/4/2024	South	Keyboard	Carol Williams	25	\$150.00	\$3,000.00	0%	\$3,000.00
4	1/5/2024	West	Mouse	David Brown	30	\$25.00	\$600.00	0%	\$540.00
5	1/5/2024	West	Mouse	Eve Davis			\$4,800.00	0%	\$4,800.00
6	1/5/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
7	1/5/2024	West	Monitor	David Brown			\$3,800.00	10%	\$3,420.00
8	1/7/2024	West	Monitor	David Brown			\$8,800.00	10%	\$7,920.00
9	1/7/2024	West	Monitor	Bob Smith			\$9,000.00	10%	\$8,100.00
10	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
11	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
12	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
13	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
14	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
15	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
16	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
17	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
18	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
19	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
20	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
21	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
22	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
23	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
24	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
25	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
26	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
27	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
28	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
29	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
30	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
31	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
32	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
33	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
34	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
35	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00
36	1/7/2024	West	Monitor	Bob Smith			\$3,800.00	10%	\$3,420.00

Dataset: industrial_output.csv
Date Range: 2000-01-01 to 2023-12-31

	REGION	OUTPUT_INDEX	EMPLOYMENT
2000-01-01	North	102.3	5021
2000-02-01	North	103.1	5033
2000-03-01	North	101.8	4987
2000-04-01	North	NA	5078
2000-05-01	North	105.2	5129
2000-06-01	North	106.0	5150
2000-07-01	North	107.3	5183
2000-08-01	North	107.3	5183
2000-09-01	North	108.1	5201
2000-10-01	North	NA	5230
2000-11-01	North	110.5	5265
2000-12-01	North	111.2	5280
2001-01-01	North	112.4	5312
2001-02-01	North	113.0	5338
2001-03-01	North	NA	5376
2001-04-01	North	115.6	
2001-05-01	North	116.4	
2001-06-01	North	116.4	
2001-07-01	North	117.1	
2001-08-01	North	NA	

Data cleaning & preprocessing

Goal: reliable inputs for analysis and modelling.

Missing values

Imputed using mean for continuous variables (Sleep Hours, Screen Time) and mode for categorical labels. Where missingness was large, variables documented and treated cautiously.

Duplicates & outliers

Exact-duplicate rows removed. Extreme outliers winsorised or inspected case-by-case to preserve signal without distortion.

Feature engineering

Created binary high-risk flags (Anxiety > threshold, Depression > threshold), standardised score z-scores, grouped screen time into bands.

Exploratory data analysis — summary

Computed central tendencies, spread, and class balances to understand population distribution and measurement characteristics.

☐ Numerical summaries

Mean and median Sleep Hours, Screen Time, Stress Level; standard deviations and IQRs used to identify dispersion.

☐ Categorical overview

Gender distribution, prevalence of high-anxiety and high-depression flags, academic performance categories.





Key visualisations

- Histogram: Age – shows modal ages and spread.
- Pie chart: Gender distribution – proportion of reported genders.
- Bar chart: Stress Level categories – frequency by level.

Visuals were generated with Matplotlib for static reporting and Plotly for interactive exploration.

Mental-health indicators

Anxiety & Depression

Distributions examined with density plots and boxplots.
Prevalence of clinically-relevant scores quantified via thresholds.

Relationship

Correlation coefficient calculated between anxiety and depression scores; scatter and regression line used to illustrate co-occurrence and strength of association.



Lifestyle factors vs mental health



Sleep Hours vs Stress

Lower sleep hours correlate with higher stress scores. Median stress is notably higher for <6 hours sleepers.



Screen Time vs Anxiety

Higher daily screen time bands associate with elevated anxiety percentiles; effect stronger after 3+ hours/day.



Physical Activity vs Depression

Regular physical activity linked to lower median depression scores; dose-response pattern observed across activity frequency.

Correlation analysis & highlights

A Pearson correlation heatmap summarises pairwise relationships. Key relationships documented for stakeholders.

Strong positive

Anxiety ↔

Depression — consistent strong positive correlation indicating comorbidity patterns.

Moderate associations

Sleep Hours negatively correlated with Stress and Anxiety. Screen Time positively correlated with Stress and Anxiety.

Academic performance

Higher stress and depression scores associated with lower academic performance on average; effect size modest but consistent.



Insights, recommendations & next steps

Key insights

Sleep deficit, excessive screen time and low physical activity are recurring correlates of worse mental-health scores. Anxiety and depression co-occur frequently.

Recommendations

- Promote sleep hygiene programmes in schools.
- Limit late-night screen exposure; introduce digital-curfew campaigns.
- Scale accessible physical-activity initiatives and group exercise.
- Expand counselling availability and targeted screening for high-risk students.

Next steps

Share interactive Plotly dashboards with stakeholders, validate thresholds with clinicians, and evaluate interventions with a pre/post study design.

- ❏ Theme colours for emphasis: #FF7047, #FFF0ED, #FFA647, #FFCE47, #ff9b7e. Use the first (#FF7047) for calls-to-action and highlight boxes.

